



CALICUT UNIVERSITY – FOUR-YEAR UNDER GRADUATE PROGRAMME (CU-FYUGP)

BSc PHYSICS HONOURS

Programme	B.Sc. Physics Honours				
Course Title	SPACE PHYSICS				
Type of Course	Elective in Major				
Semester	VI				
Academic Level	300 - 399				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	4	-	-	60
Pre-requisites	PHY4CJ205- Modern Physics				
Course Summary	This course introduces the student to Space Physics. The various subdisciplines of the topic such as structure and properties of the solar system with emphasis on Earth and the Sun and their magnetic fields, the elements of planetary science, the rudiments of space weather as well as basics of space flight dynamics are dealt with in detail.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Understand the basic structure and parameters of the Earth and the Sun including their atmospheres and their magnetic fields	U	C	Instructor-created exams / Quiz
CO2	Understand the basic elements of planetary science including the structure of the solar system and the classification of its constituents	U	C	Viva Voce / Home Assignments/ Seminar Presentations

CO3	Understand the basics of space weather and its various phenomena such as solar wind, interplanetary space and solar activities like coronal mass ejections	U	C	Instructor-created exams / Quiz
CO4	Understand the theory behind the orbital dynamics and the technology of rocket and spacecraft propulsion	U & Ap	C	Instructor-created exams / Quiz
CO5	Interpret the complex structures and dynamics of Earth's magnetosphere, including the polar cusp, plasma sheet, ring current, radiation belts, and associated wave phenomena.	Ap	C	Instructor-created exams / Quiz
CO6	Equip with the knowledge and skills necessary to apply principles of space science in analyzing and understanding various phenomena within our solar system and beyond.	U	C	Instructor-created exams / Quiz
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (48 +12)	Marks (70)
I	The Earth and the Sun		14	20
	1	The Earth - Gross properties, Internal structure of the Earth	2	
	2	The terrestrial atmosphere, The Earth's magnetic field	2	
	3	Motions of the Earth, Solar – Terrestrial relations, The Earth in Space	2	
	4	The Sun - Introduction, Vital statistics of the Sun, The Solar Photosphere,	3	
	5	Structure of Solar atmosphere, The solar interior, Sunspots and solar cycle	3	
	6	Other features of Solar activity, Radio Studies of the Quiet Sun, Radio radiation of the disturbed Sun	2	

	Sections 5.1 to 5.6,5.8, 4.1 -to 4.3, 4.5-4.10 of Book 1			
II	Planetary Science		14	20
	7	Planetary science – Introduction, the solar system in the last four millennia	2	
	8	Origin of the solar system, Evolution of atmospheres	3	
	9	Terrestrial planets, Outer planets, structure and classification	3	
	10	Comets, Asteroids, their origin and properties	2	
	11	Magnetospheres of planets, their structure and prominent properties	2	
	12	Planetary missions, Other solar systems	2	
	Sections 3.1 to 3.11 of Book 2			
III	Space Weather		10	15
	13	What is a space plasma, What is a plasma, The realm of plasma physics	2	
	14	The solar wind and interplanetary magnetic field, Magnetic reconnection	2	
	15	Space weather – introduction, Solar activity, The Solar wind	2	
	16	Aurora, Auroral substorms, Solar flares, The ionosphere	2	
	17	Coronal mass ejections and geomagnetic storms, Magnetic storms and substorms	2	
	Sections 5.1 to 5.5, 5.7, 5.8, 5.11, 5.14 of Book 2			
IV	Orbital Dynamics		10	15
	18	Celestial Mechanics - Foundations, Attraction of a spherical body	1	
	19	The two – body approximation, The two – body orbit, Kepler's equation, Determination of orbit	2	
	20	Space Dynamics - The energy requirements, Rocket propulsion	2	
	21	Sub – orbit flights, Artificial satellites, Lunar and Planetary probes	3	
	22	Multistage rockets- introduction, Reusable launch vehicles	2	
	Sections 2.1 to 2.9, 3.1 to 3.6 of Book 1 Sections 7.1, 7.4,7.7 of Book 3			

V	Open Ended Module: The Terrestrial Magnetosphere		12	
		he structure of the Magnetosphere The polar cusp, The near - Earth plasma sheet, The ring current The plasmasphere, The radiation belts; the South Atlantic anomaly Waves in the magnetosphere. Classification of waves		
	References: Books 2-5			

Books and References:

1. Astrophysics of the Solar System – K D Abhyankar, Universities Press, 1999 (Book 1)
2. Space Science – Louise K. Harra & Keith O. Mason, Imperial College Press, London, 2004 (Book 2)
3. Space Flight Dynamics – William E Wiesel, McGraw Hill, 3rd Ed., 2010 (Book 3)
4. Space Physics: An Introduction – C T Russell, Luhmann & Strangeway, Cambridge University Press, 2016
5. Astrophysics, Stars and Galaxies - K D Abhyankar, Universities Press, 2001
6. A Question and Answer Guide to Astronomy by Bely, Christian and Roy, Cambridge University Press
7. Introduction to Space Physics – M.G. Kivelson & C.T.Russell, Cambridge University Press, 1995

Mapping of COs with PSOs and POs :

	PSO 1	PSO 2	PSO 3	PSO 4	PS O5	PS O6	PO1	PO2	PO3	PO4	PO5	PO 6	PO 7
CO 1	3	0	0	0	0	0	3	2	0	0	0	0	3
CO 2	3	2	0	0	0	0	3	2	0	0	0	0	3
CO 3	3	2	0	0	0	0	3	2	0	0	0	0	3
CO 4	3	0	0	2	0	0	3	2	0	2	0	0	3
CO 5	3	0	0	0	0	0	3	2	0	0	0	0	3
CO 6	3	0	0	0	0	0	3	2	0	0	0	0	3

Correlation Levels:

Level	Correlation
0	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Discussion / Seminar
- Internal Theory/Practical Exam
- Assignments /Viva
- End Semester Exam (70%)

Mapping of COs to Assessment Rubrics

	Internal Theory/ Practical Exam	Assignme nt /Viva	Practical Skill Evaluation	End Semester Examinations
CO 1	✓	✓		✓
CO 2	✓	✓		✓
CO 3	✓	✓		✓
CO 4	✓	✓		✓
CO 5	✓	✓		✓
CO 6		✓	✓	